

18 February 2020

وَمَا أُوتِيتُمْ مِنَ الْعِلْمِ إِلَّا قَلِيلًا

24 جمادى الثانية 1441

Analog IC Design

Lecture 03

Review on Semiconductors Basics

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Outline

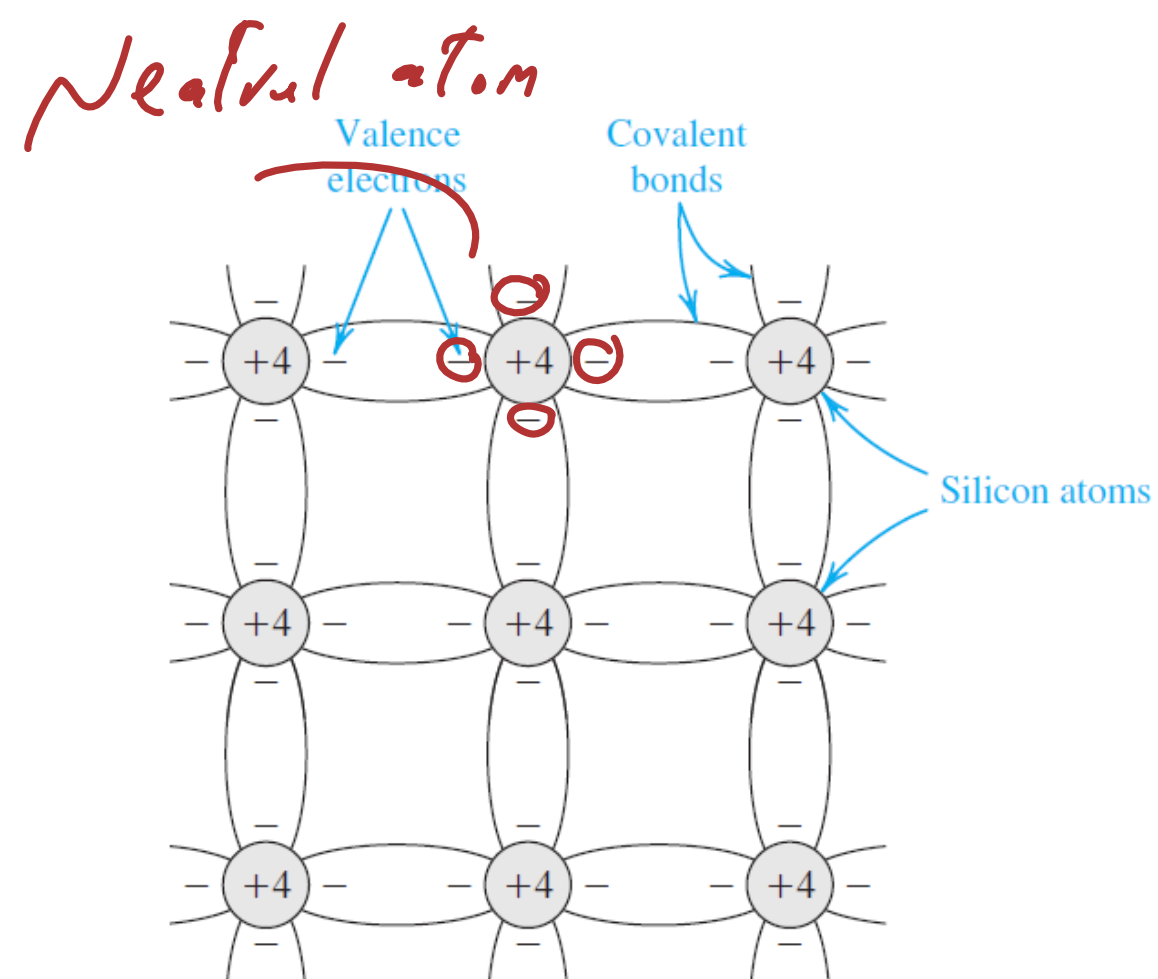
- ☐ What are semiconductors?
- ☐ Electrons and holes
- ☐ N-type and P-type silicon
- ☐ Drift and diffusion current
- ☐ The PN-junction

What are Semiconductors

- ☐ Conductors → Ex: copper
- ☐ Insulators → Ex: glass
- ☐ Semiconductors are materials whose conductivity lies between that of conductors and insulators
- ☐ **What is so special about semiconductors?**
 - The electrical conductivity can be dramatically changed by introducing extrinsic dopant atoms *doping / sub*
 - We have two types of carriers: electrons and holes
- ☐ Silicon (Si) is the semiconductor material used in the majority of today's electronic devices

Silicon Crystal

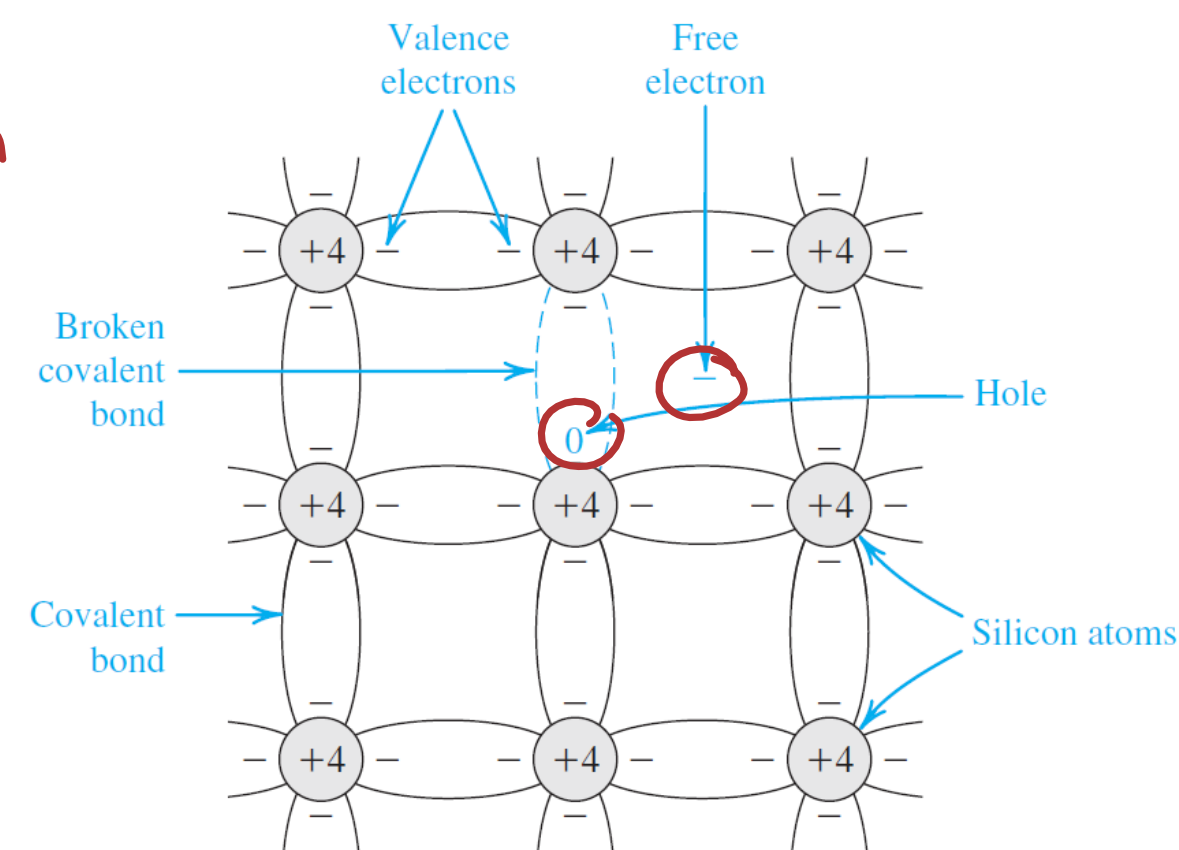
- ❑ Covalent bonds are formed by sharing of the valence electrons
- ❑ At 0 K, all bonds are intact and no free electrons are available



Electrons and Holes

- ❑ At room temperature, some of the covalent bonds are broken by thermal generation
- ❑ Each broken bond gives rise to a free electron (e^-) and a hole (h^+)
 - Both e^- and h^+ become available for current conduction

Conduction
in valence
band



Intrinsic Silicon

- ❑ Carrier concentration is the number of charge carriers per unit volume (cm^3)
- ❑ At thermal equilibrium, the recombination rate is equal to the generation rate
- ❑ The concentration of free electrons (n) is equal to the concentration of holes (p)
 $n = p = n_i$ intrinsic silicon
- ❑ The product of n and p is constant (depends only on temperature)
 $np = n_i^2$ Math action law

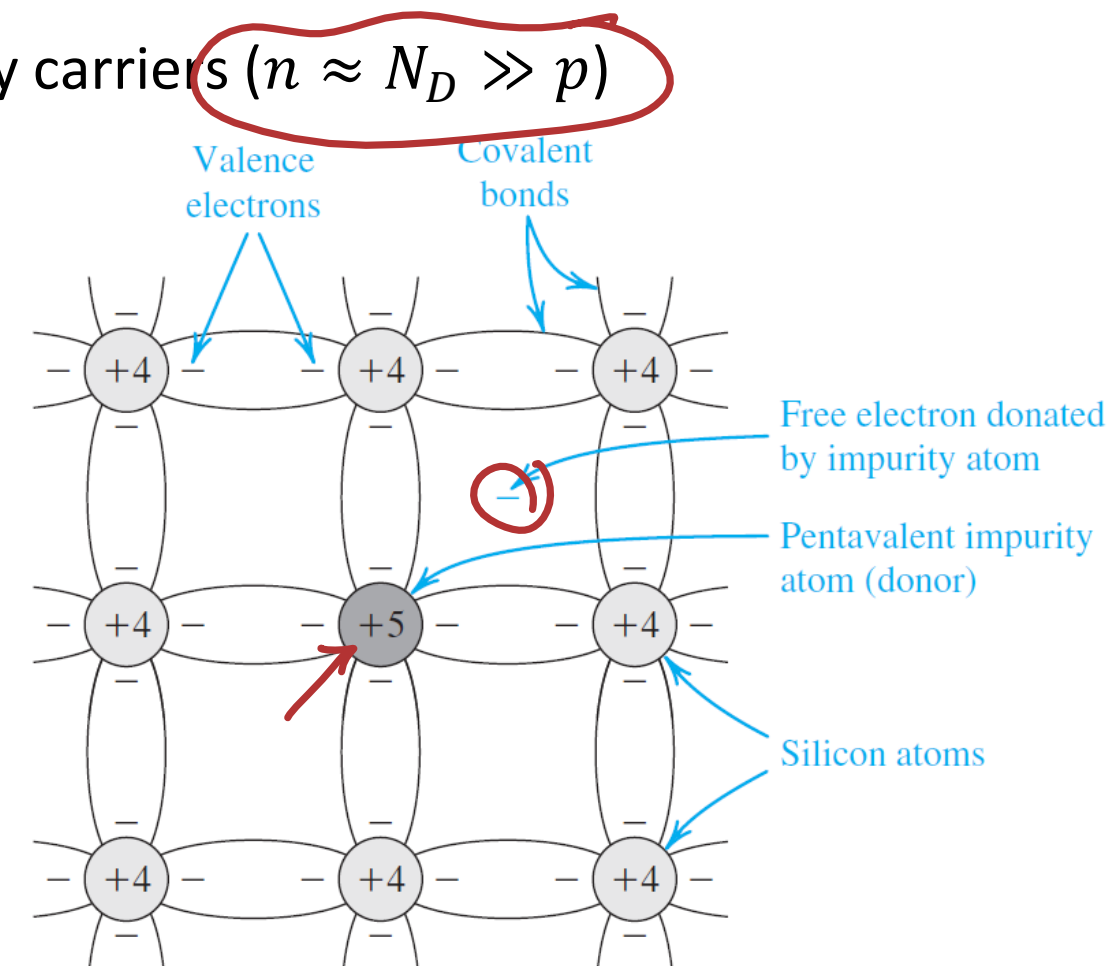
Doped (Extrinsic) Silicon *Impurities*

- ❑ Doping involves introducing impurity atoms into the silicon crystal
- ❑ To increase the concentration of free electrons (n) silicon is doped with a pentavalent (valence = 5) impurity (Ex: Phosphorus)
 - Each dopant atom (donor) gives a free e^- and a fixed positive charge (+ve ion) $\bar{e}$ *نقدور*
 - Electrons become the majority carriers ($n \gg p$)
 - The doped silicon is n-type *Majority*
- ❑ To increase the concentration of holes (p) silicon is doped with a trivalent (valence = 3) impurity (Ex: Boron)
 - Each dopant atom (acceptor) gives a h^+ and a fixed negative charge (-ve ion)
 - Holes become the majority carriers ($p \gg n$)
 - The doped silicon is p-type *Majority*

$$n \cdot p = n_i^2$$

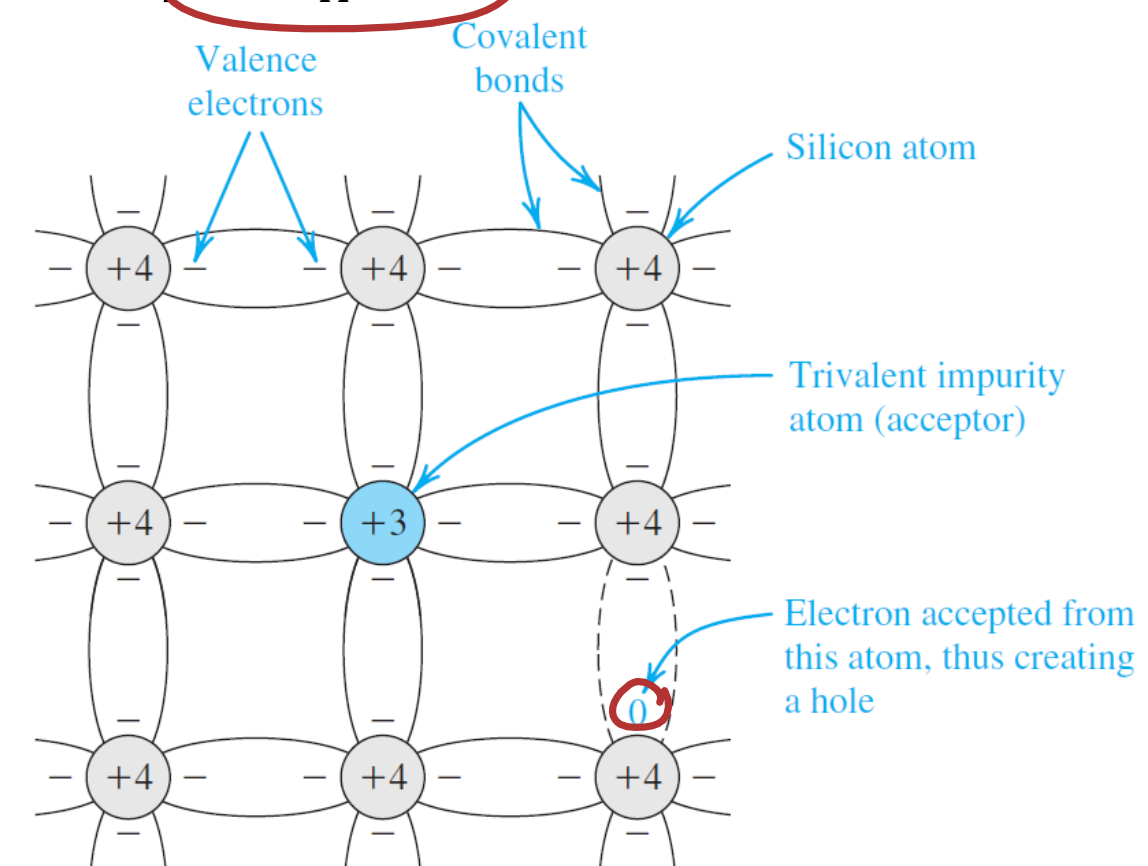
N-Type Silicon

- ❑ Each dopant atom (donor) gives a free e^- and a fixed positive charge (+ve ion)
 - N_D : donor concentration
- ❑ Electrons become the majority carriers ($n \approx N_D \gg p$)



P-Type Silicon

- ❑ Each dopant atom (acceptor) gives a h^+ and a fixed negative charge (-ve ion)
 - N_A : acceptor concentration
- ❑ Holes become the majority carriers ($p \approx N_A \gg n$)



Current Flow: (1) Drift Current

- Current flows due to electrical field (E)

- Holes are accelerated in the direction of E

$$h^+ \text{ drift velocity} = v_{p\text{-drift}} = \mu_p E$$

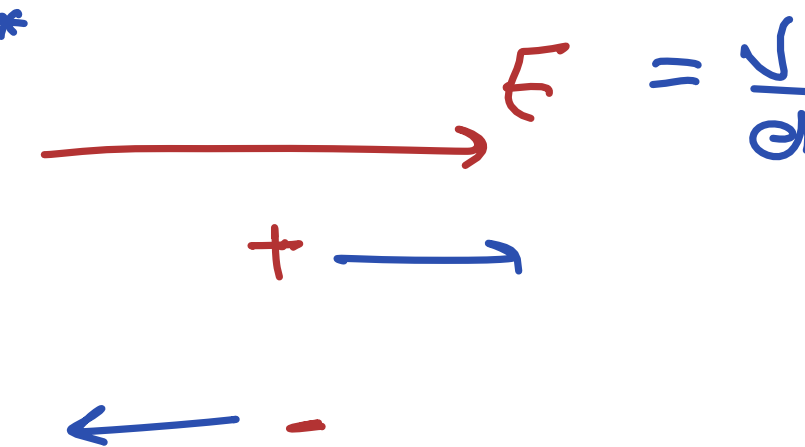
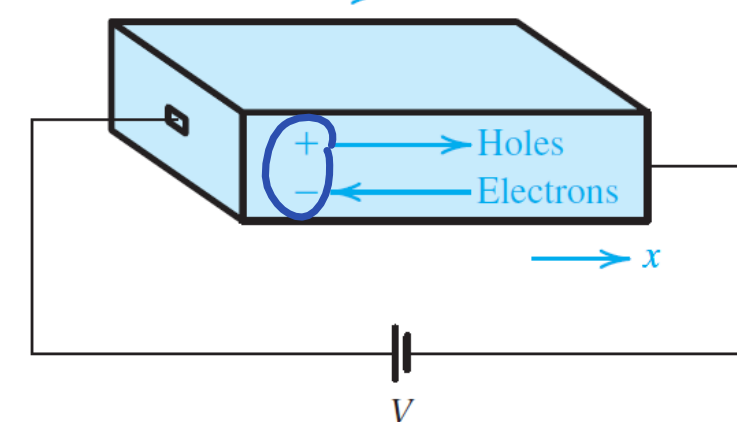
- Free electrons are accelerated in the direction opposite to E

$$e^- \text{ drift velocity} = v_{n\text{-drift}} = -\mu_n E$$

- μ is the mobility: $\mu_n = 2 - 4$ times μ_p

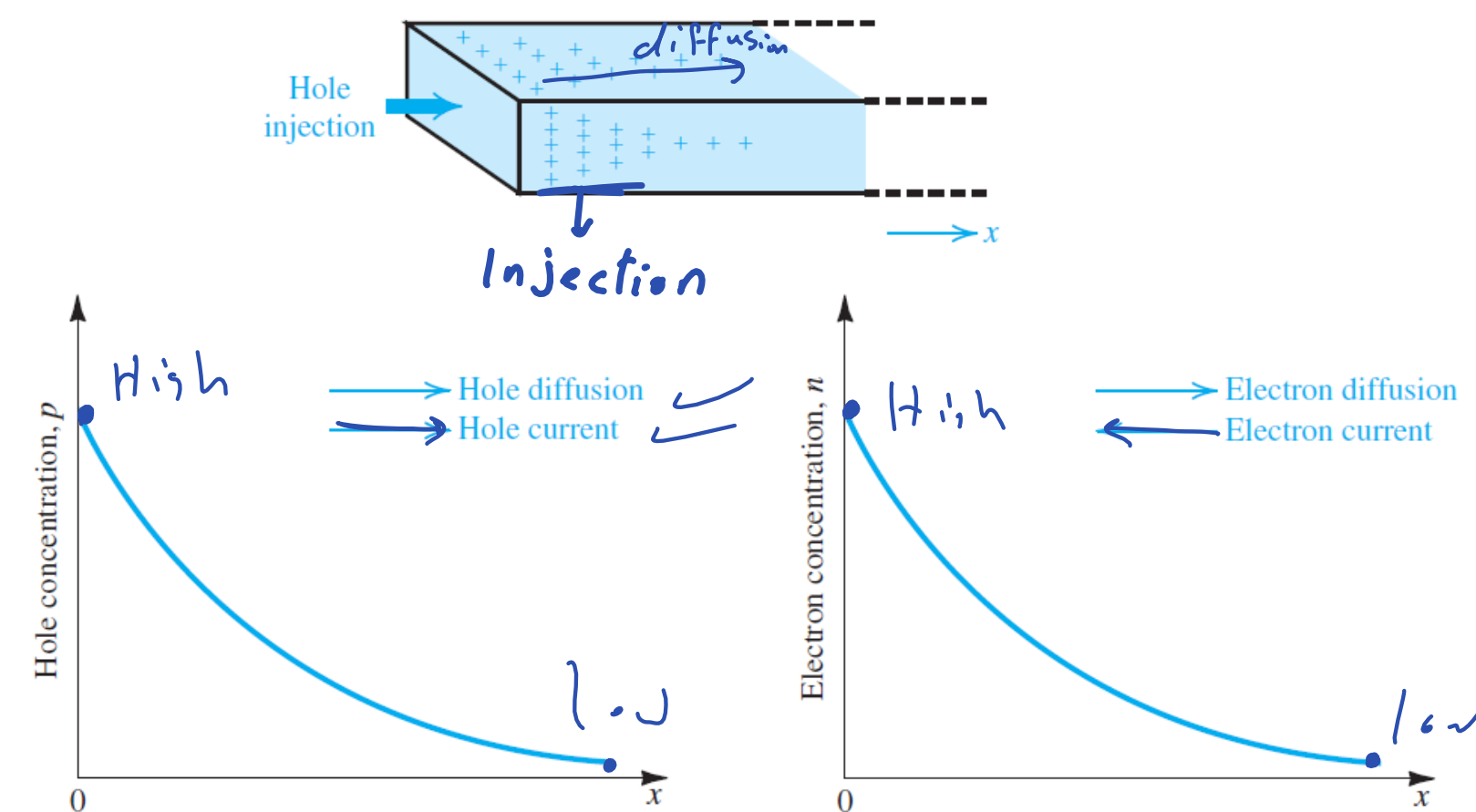
- Note that if there are no carriers, there will be no current, even if there is an electric field

* وجود E لا يضمن وجود التيار (current)
لازم يبقوا حوامل (carriers)

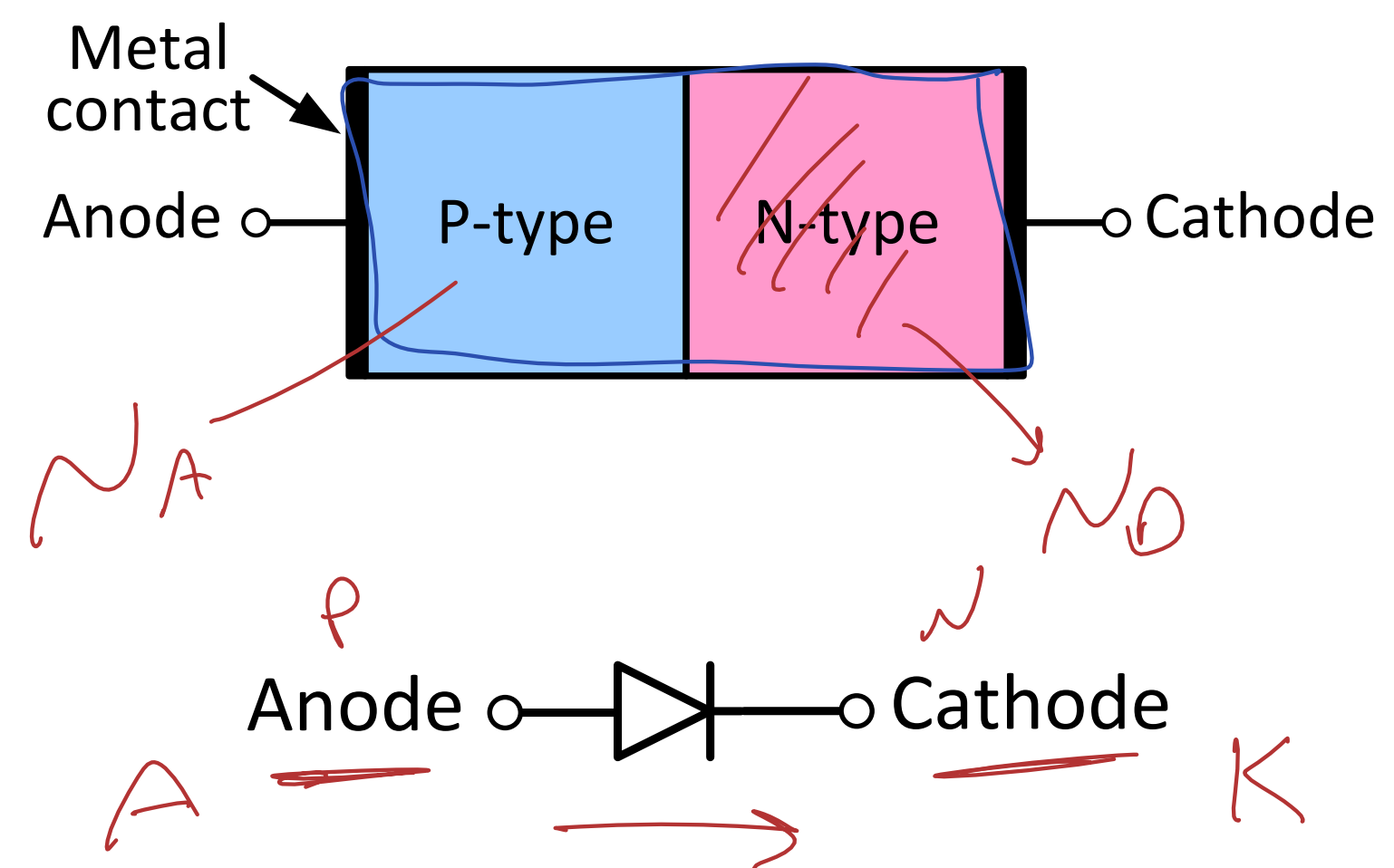


Current Flow: (2) Diffusion Current

- Current flows due to carrier concentration gradient
 - Carriers diffuse from the region of high concentration to the region of low concentration

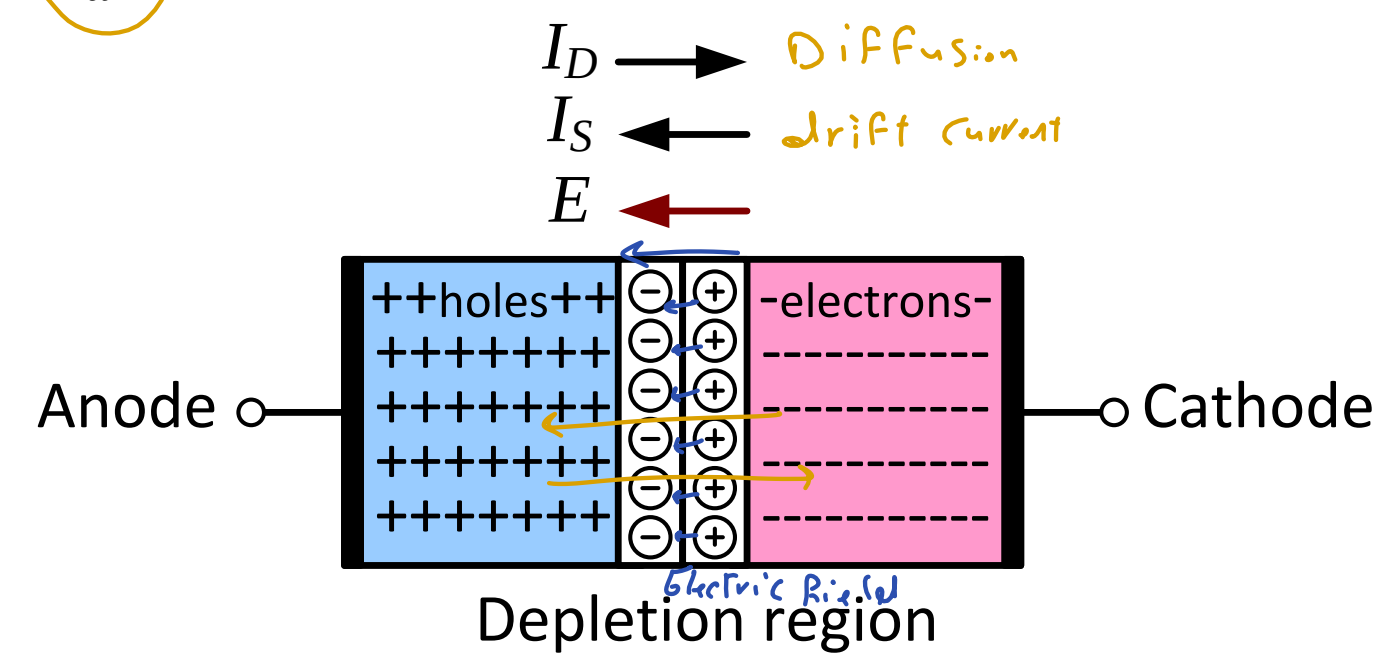


The PN Junction (The Diode)



PN Junction in Equilibrium (o.c.)

- ❑ Diffusion current (I_D) flows due to concentration gradient
 - A depletion region of uncovered fixed charges is formed
 - The uncovered charges create $E \rightarrow$ drift current (I_S)
- ❑ $I_D = I_S \rightarrow$ net current ($I_D - I_S$) is zero
- ❑ Capacitance (C) $= \frac{\epsilon A}{d}$



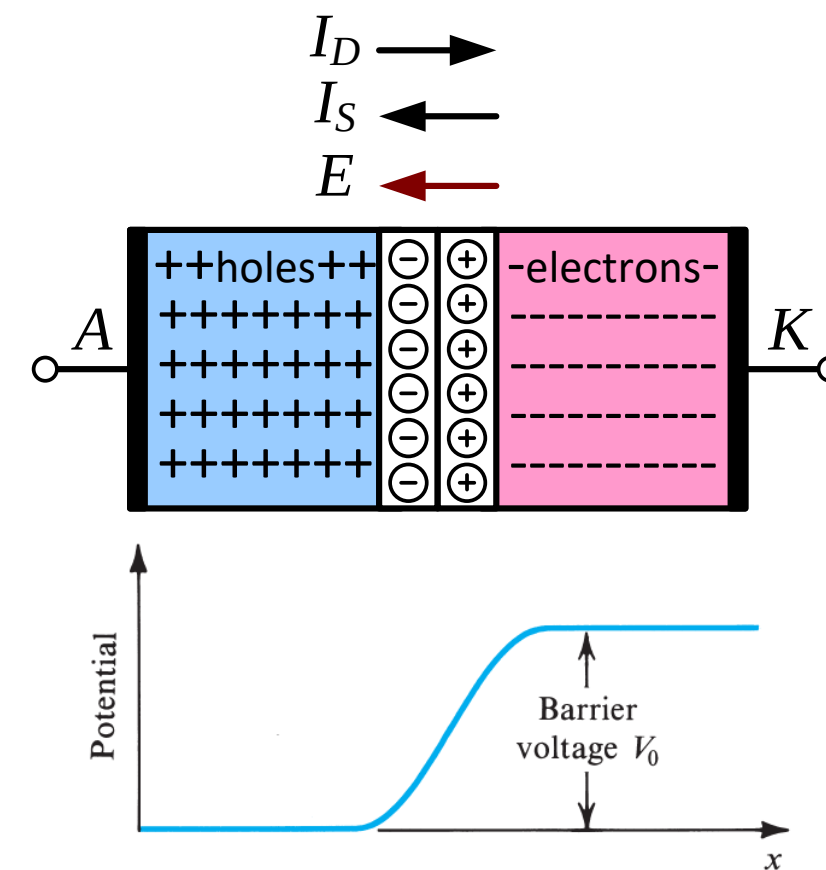
A acceptor atoms
 -ve fixed charge
 holes diffuse
 Fixed charges
 holes
 Donor atoms
 electrons

PN Junction in Equilibrium (o.c.)

- Built-in electric field and barrier voltage due to depletion region

$$V_0 = V_T \ln \left(\frac{N_A N_D}{n_i^2} \right) \approx 2.3 V_T \log \left(\frac{N_A N_D}{n_i^2} \right) \approx 0.6 - 0.9V$$

- The barrier voltage (V_0) limits carrier diffusion

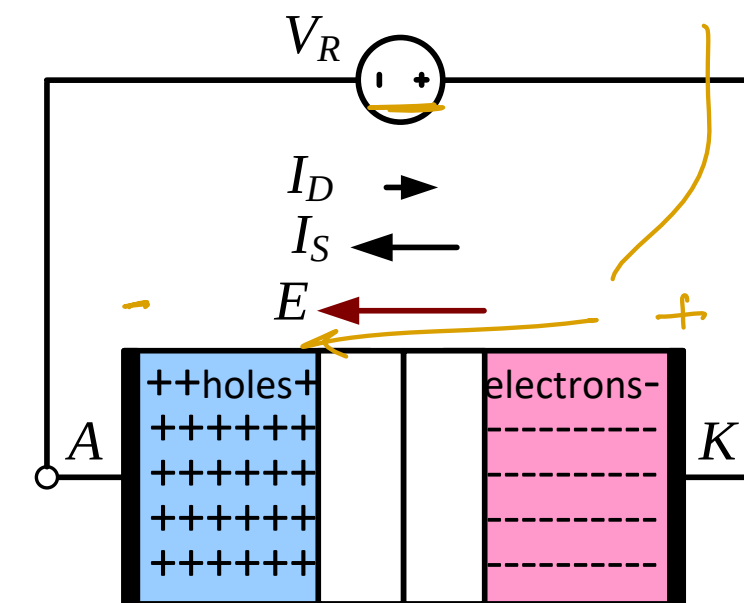
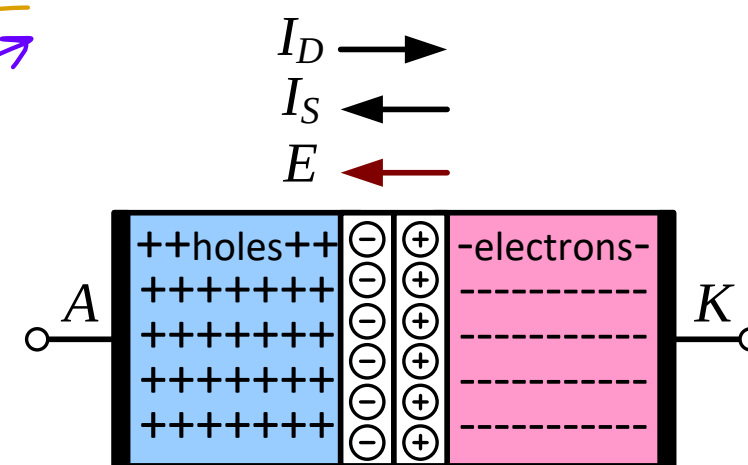


PN Junction in Reverse (Rvr) Bias

- ❑ The applied reverse voltage increases diffusion barrier
 - Opposes diffusion current
- ❑ Electric field increases ✓✓
 - But drift current almost unchanged: no carriers to accelerate
- ❑ Net current is very small $\approx -I_S$
- ❑ Depletion width increases → capacitance decreases
 - Capacitance at zero bias is larger

d T

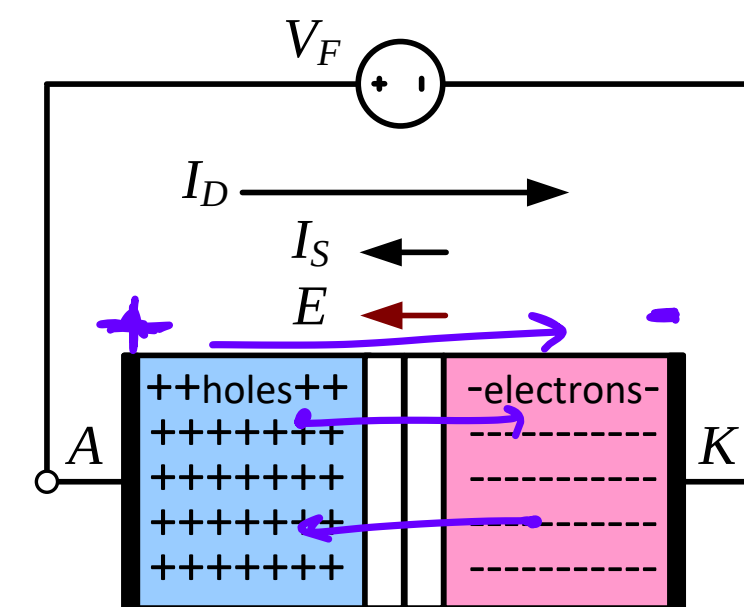
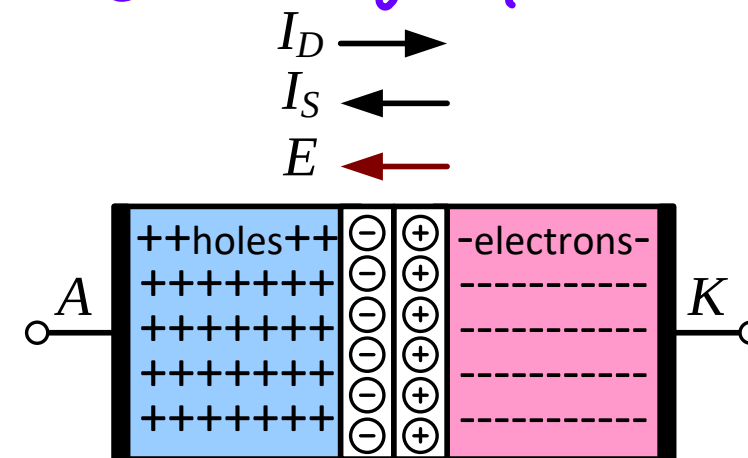
$$C = \frac{\epsilon A}{d}$$



PN Junction in Forward (Fwd) Bias

- ❑ The applied forward voltage **decreases diffusion barrier**
 - Dramatically increases diffusion current
- ❑ Net current is very high = $I_D - I_S \approx I_D = I_S e^{\frac{V_F}{V_T}}$
 - Forward current exponentially increases with voltage across diode (V_F)
- ❑ Depletion width decreases

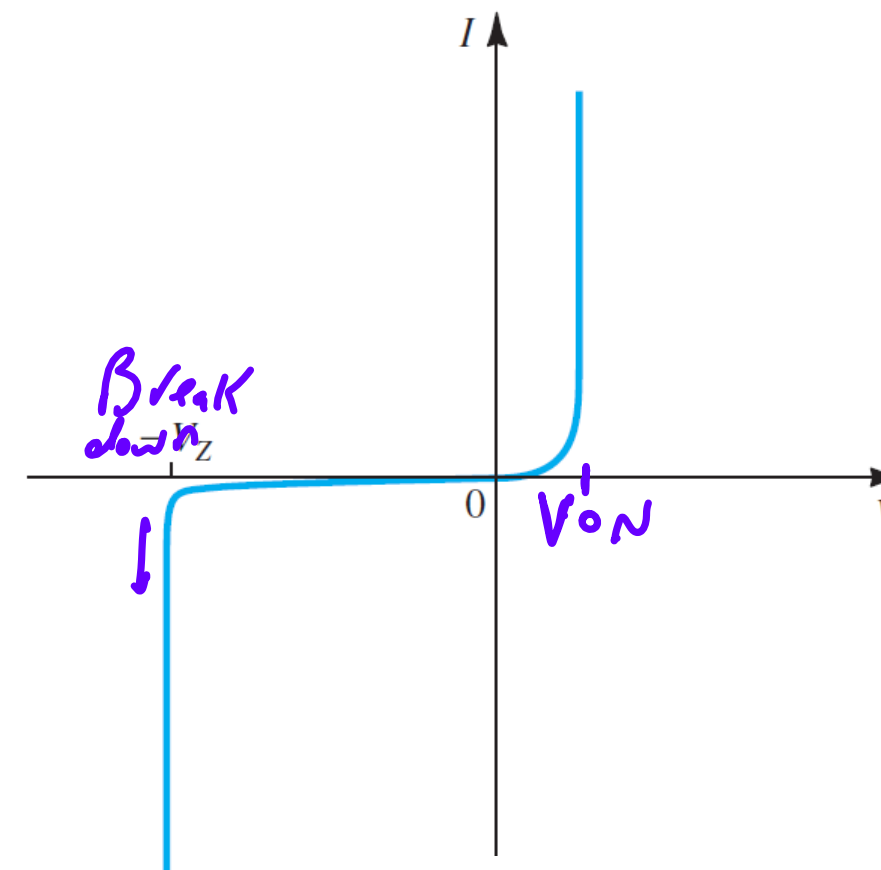
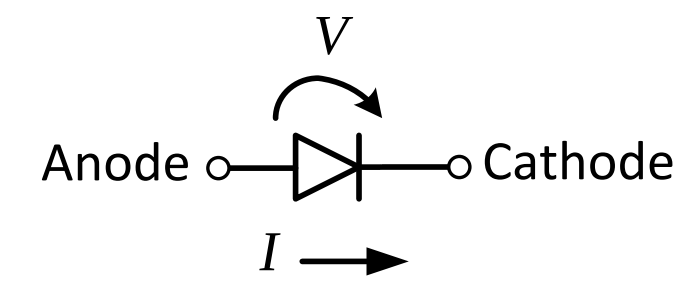
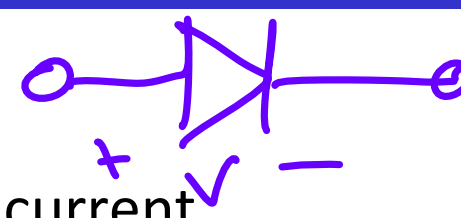
* پس بجا آنکه در Current بگیر ما پس میوه را
 / به عنوان دیود که پوی



PN Junction IV Characteristics

☐ $I = I_S(e^{\frac{V}{V_T}} - 1)$

- ☐ Forward: High diffusion current exponentially dependent on $V = V_F$
- ☐ Reverse: Very small drift current almost independent of $V = -V_R$
- ☐ Breakdown: Very high reverse current at LARGE reverse bias voltage



[Sedra/Smith, 2015]

Thank you!

References

- ❑ A. Sedra and K. Smith, “Microelectronic Circuits,” Oxford University Press, 7th ed., 2015.
- ❑ B. Razavi, “Fundamentals of Microelectronics,” Wiley, 2nd ed., 2014.